

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ
— IN THE NAME OF ALLAH —



الجامعة الإسلامية العالمية شيتاغونغ
International Islamic University Chittagong

Dept. of Computer Science & Engineering

Semester: 5th

Section: 5AF

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Course Title: Electrical Drives and Instrumentation Lab

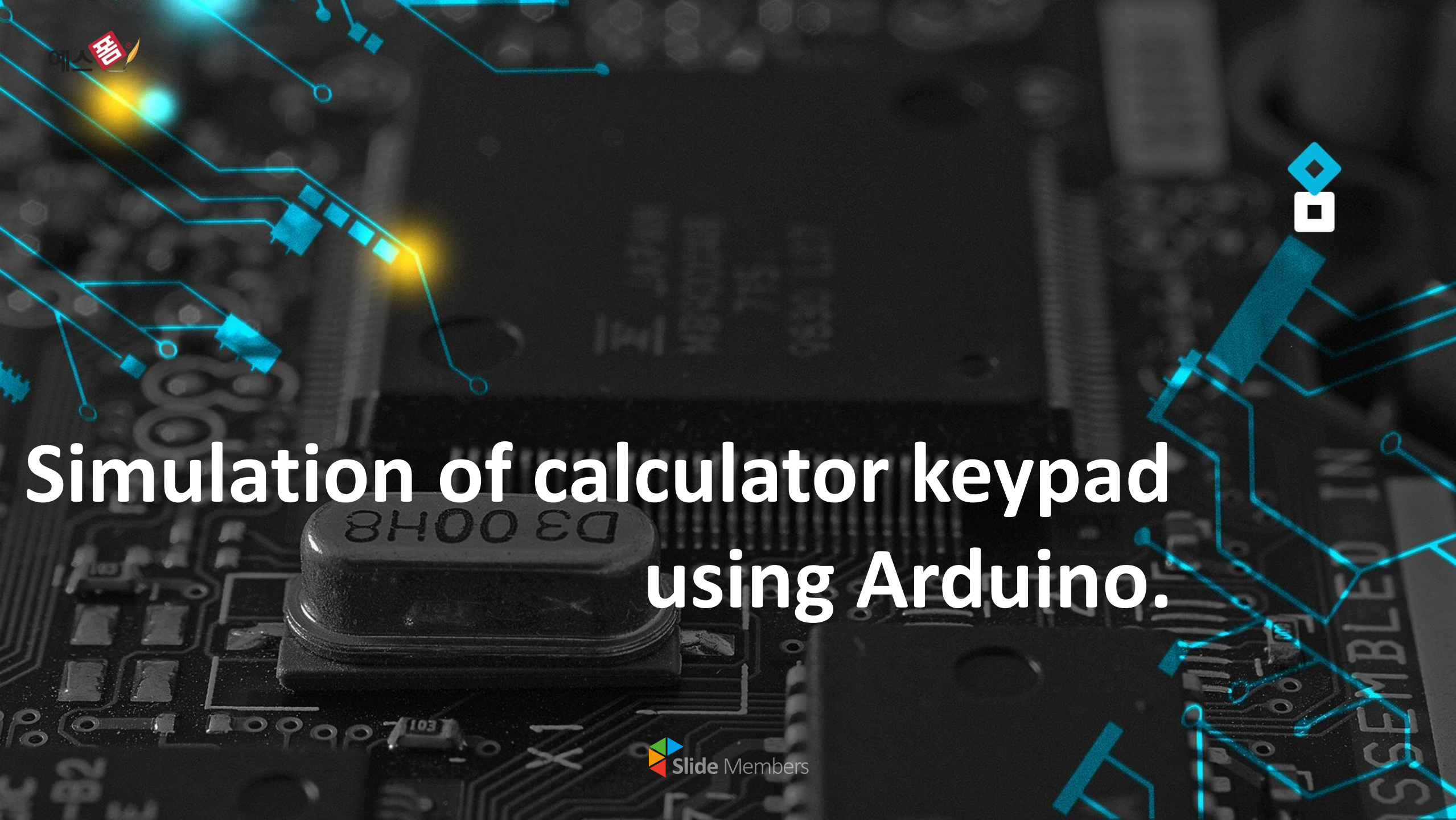


GROUP MEMBER

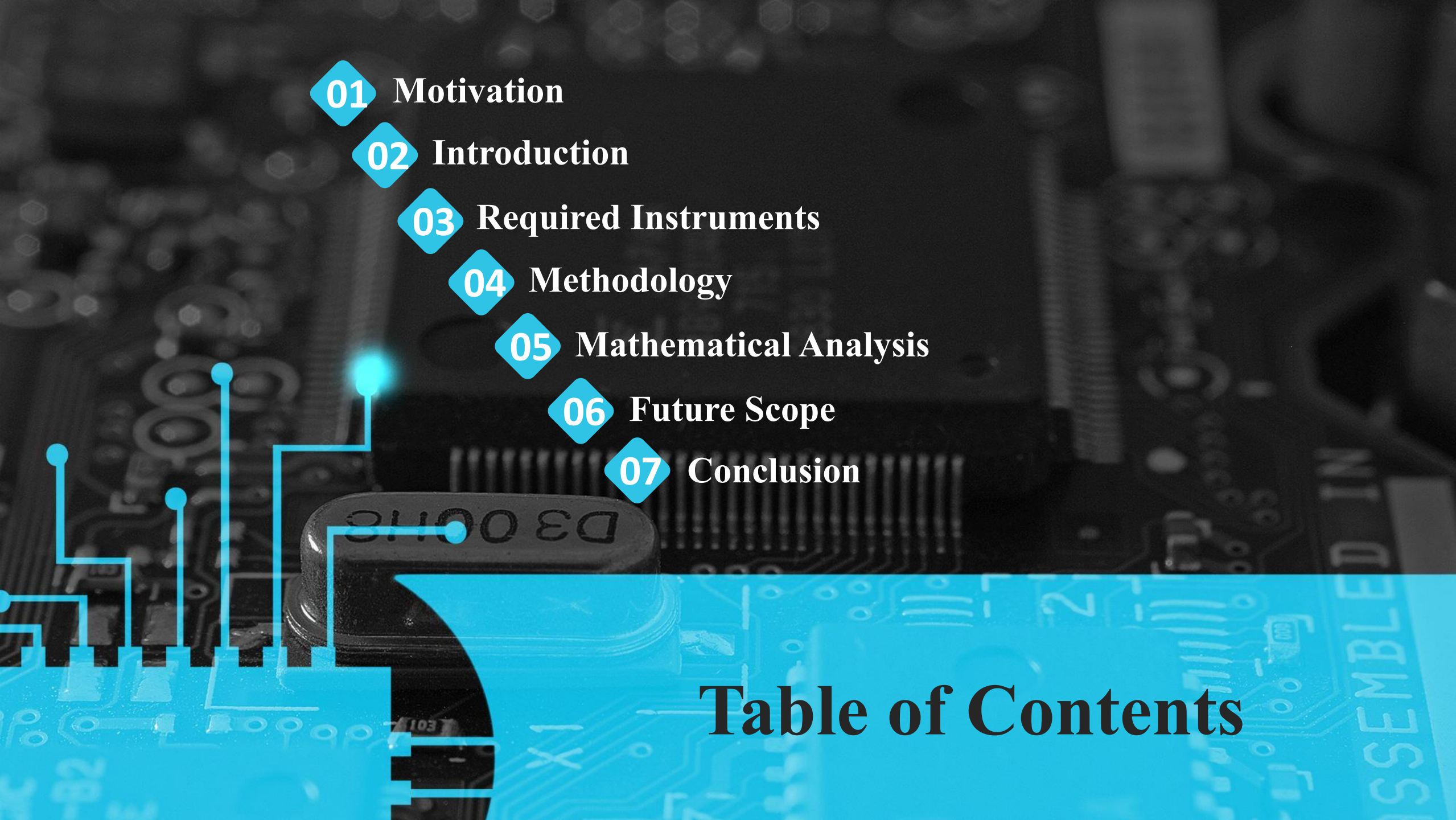
	<u>Submitted By:</u> C201203 - Faria Sultana Avir C201213 - Faiza Binta Ali C201225 - Khadizatul Kubra C201233 - Sadia Elias C201242 - Farida Nusrat

COURSE TEACHER

	<u>Submitted To:</u> MD Belal Uddin Sifat Adjunct Faculty.



Simulation of calculator keypad using Arduino.



01 Motivation

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01#

Motivation

Motivation

- A calculator is an essential thing that we use in our day-to-day life. We wanted to create a calculator as it is so much necessary for a student.
- We are using calculator since we were in school. Since then, we were curious to know the function of a calculator. That curiosity leads us to create an Arduino-based Calculator.





02#

Introduction

Introduction

A calculator is a small, portable electronic device used to perform both basic operations of arithmetic.. In this project, we have Designed Arduino calculator for calculating simple arithmetic operations using Arduino Uno. This Arduino calculator will calculate arithmetic operations such as addition, subtraction, multiplication, division, etc...



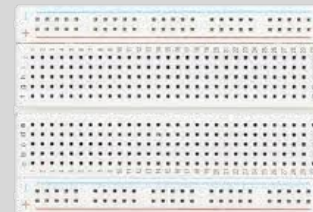


03#

Required Instrumentation

Required Instrumentation

- **Arduino Uno**
- **4*4 Keypad**
- **Bread Board**
- **16*2 LCD with pin**
- **Jumped Wires**
- **Arduino Cable**
- **9V Battery**



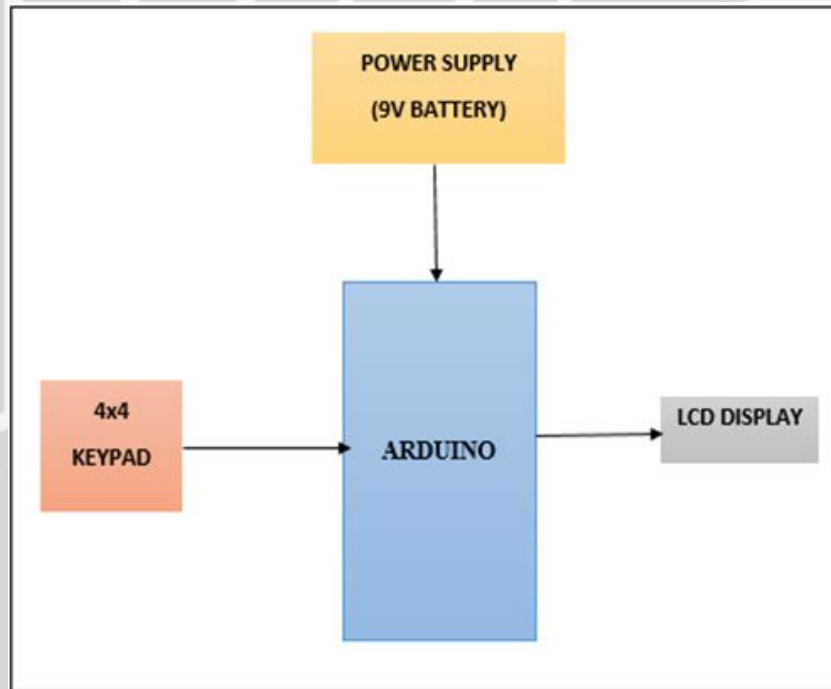


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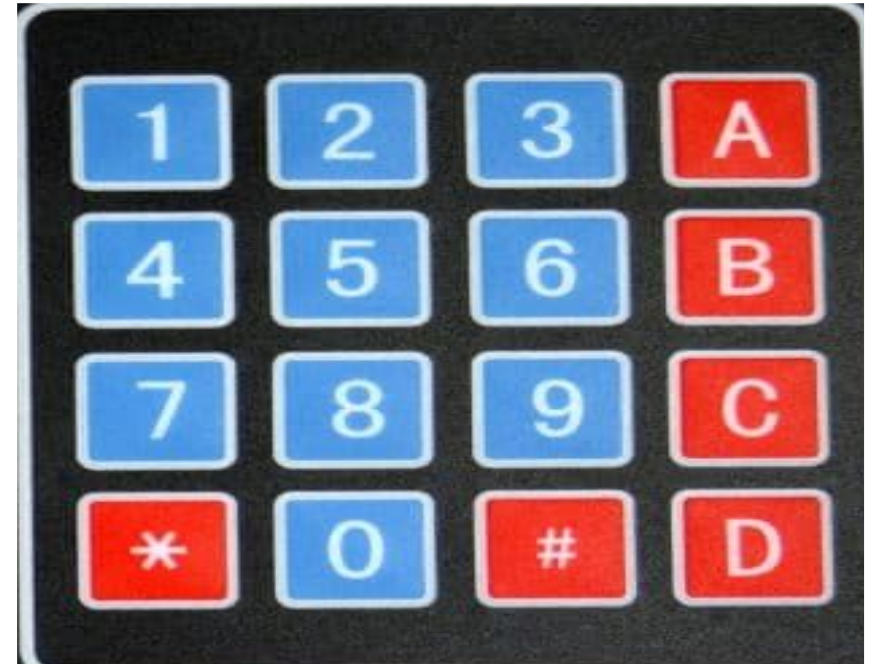
Methodology

Methodology

❑ Block Diagram



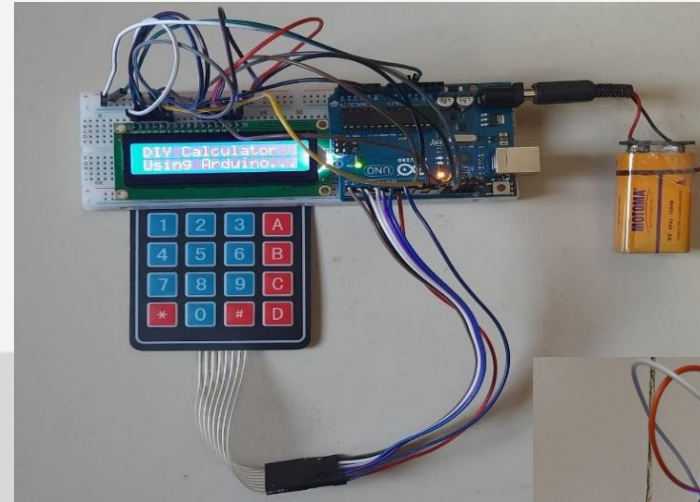
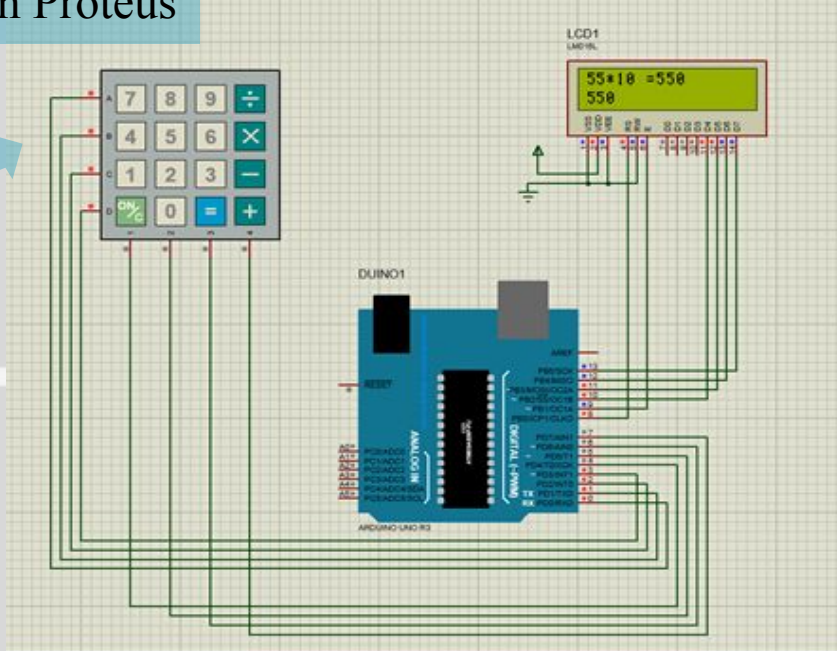
4x4 Keypad



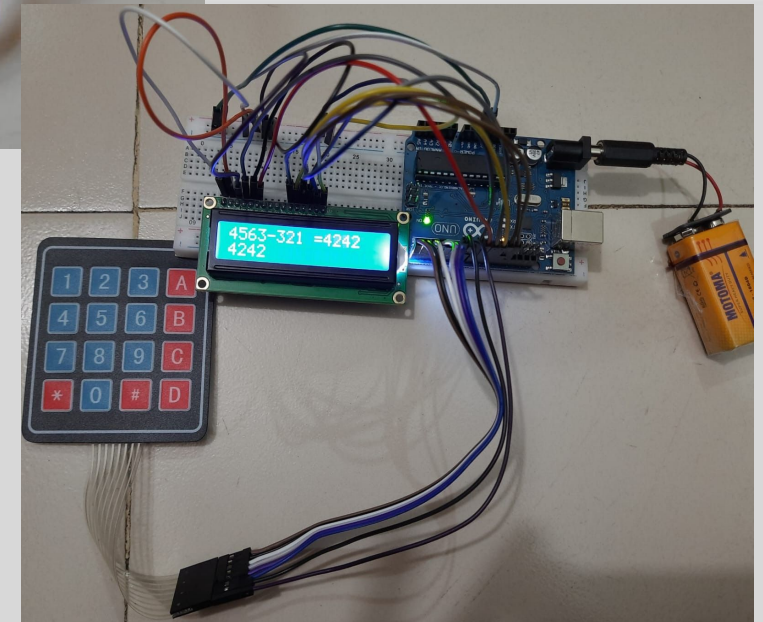
Methodology

❑ Circuit Diagram

Implement in Proteus

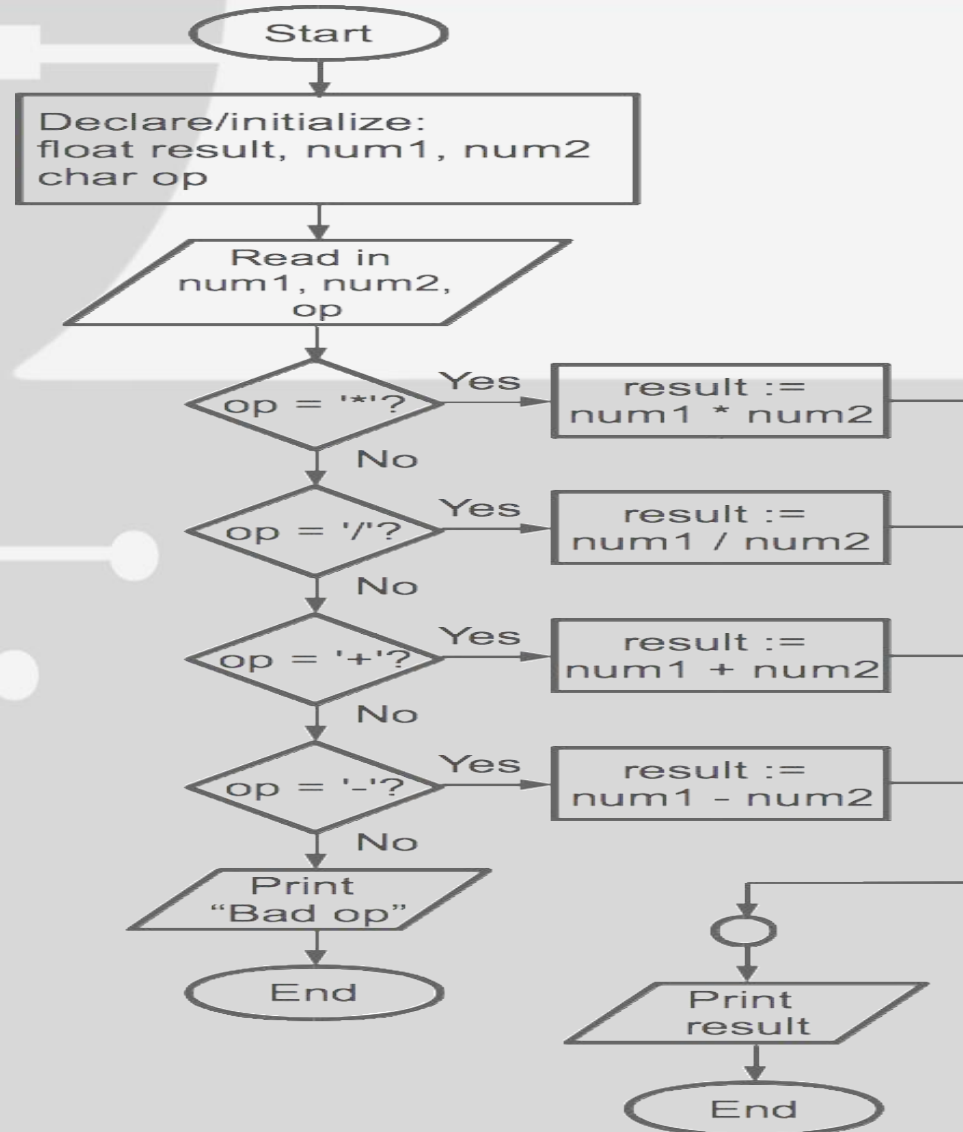


Implement Pictures



Methodology

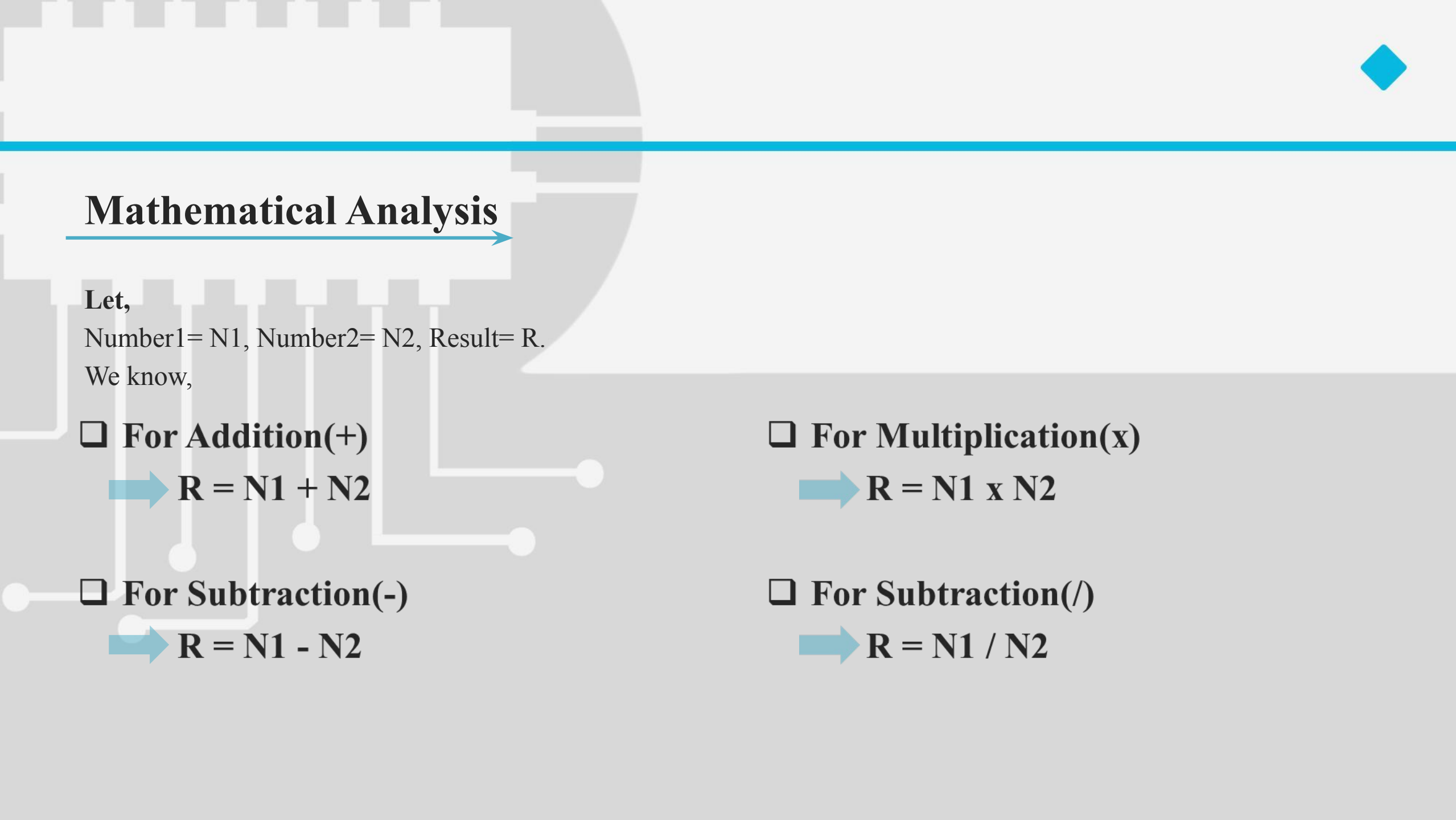
□ Flowchart





05#

Mathematical Analysis



Mathematical Analysis

Let,

Number1= N1, Number2= N2, Result= R.

We know,

❑ For Addition(+)

➡ $R = N1 + N2$

❑ For Subtraction(-)

➡ $R = N1 - N2$

❑ For Multiplication(x)

➡ $R = N1 \times N2$

❑ For Subtraction(/)

➡ $R = N1 / N2$

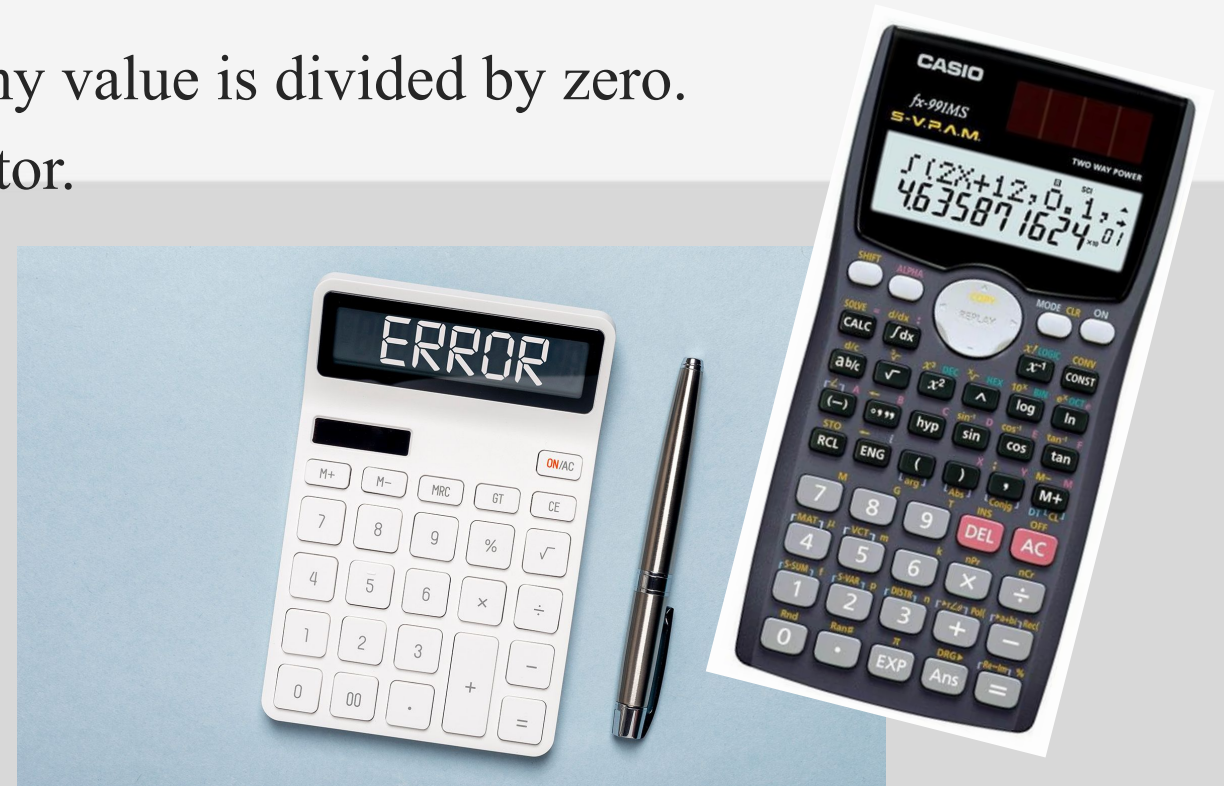


06#

Future Scope

Future Scope

- ❑ An invalid value will be shown when any value is divided by zero.
- ❑ Will be converted to a scientific calculator.





07#

Conclusion



Conclusion

- ❑ In this project, we have created a portable, reliable, and low-cost microcontroller-based calculator with a simple design for simple and fast calculation.
- ❑ This project is not only for calculations it shows us how the keypad acquires individually entered characters and numbers from all the numbers and characters.
- ❑ It also explains how Arduino controls the LCD and Keypad.
- ❑ Using this calculator we can do arithmetic calculation hundred percent accurately and the result matches with the result of other scientific calculator.

THANK YOU

